

Customer Churn Prediction & Retention Analysis Dashboard

1. Introduction

Customer churn is a major challenge for subscription-based businesses such as telecom, banking, SaaS, and streaming platforms. Losing customers directly impacts revenue, customer lifetime value, and business growth.

This project focuses on analyzing customer behavior and identifying patterns that contribute to customer churn using Power BI, Python, and Machine Learning techniques.

2. Project Objectives

- Analyze customer churn behavior
- Identify high-risk customers
- Build an interactive dashboard
- Track important business KPIs
- Support data-driven decision-making

3. Dataset Description

Dataset Used: IBM Telco Customer Churn Dataset

Features include:

- Customer demographics
- Contract details
- Monthly charges
- Internet services
- Payment methods
- Customer tenure
- Churn status

4. Tools & Technologies Used

- Power BI
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- DAX

5. Data Cleaning & Preparation

- Removed missing values
- Converted TotalCharges to numeric format
- Removed unnecessary columns
- Encoded target variable
- Handled categorical variables
- Validated data consistency

6. Key KPIs (Using DAX)

- Total Customers
- Churned Customers
- Churn Rate
- Average Monthly Charges
- Total Revenue
- Revenue Lost Due to Churn

7. Dashboard Overview

The dashboard includes:

- KPI Cards
- Churn by Contract Type
- Customer Distribution by Gender
- Churn by Internet Service
- Tenure Impact on Churn
- Payment Method Analysis
- Interactive Filters & Slicers

8. Key Insights & Findings

- Month-to-month contracts have the highest churn
- Customers without tech support churn more
- Short-tenure customers are high-risk
- Higher monthly charges increase churn
- Electronic check users have higher churn

9. Business Impact

- Improved retention strategies
- Reduced revenue loss
- Better customer experience
- Data-driven business decisions

10. Conclusion

The Customer Churn Prediction Dashboard successfully analyzes customer behavior and identifies key factors contributing to churn.

The project provides actionable insights that support customer retention and business growth.

11. Future Enhancements

- Machine Learning Integration
- Churn Probability Scoring
- Real-Time Dashboard
- Customer Segmentation
- Revenue Forecasting
- AI-Based Recommendations

12. Future Enhancements

